Product Requirement Document (PRD): MINT (Market Intelligence & Network Transfers)

1. Executive Summary

MINT (Market Intelligence & Network Transfers) is an AI-powered financial assistant designed to perform real-time market analysis, prepare trading strategies, and execute trades directly on Stellar's Decentralized Exchange (SDEX). To protect user capital, all automated trading features in MINT operate under a strict, non-bypassable spending and loss cap. Beyond automated trading, MINT provides peer-to-peer (P2P) transfers, multi-asset payments, and cross-border remittances leveraging Stellar's low-cost payment rails.

2. Core User Personas

- **Automated Crypto Trader:** Uses MINT as an AI co-pilot to scan markets, detect trading opportunities, and execute trades 24/7 without risking catastrophic loss.
- **Daily Utility User:** Uses MINT to make instant payments, exchange assets with low fees, and send peer-to-peer micro-transfers.

3. Product Architecture & Technical Stack

- **User UI:** MINT Web & Mobile Chat-First Dashboard
- **AI Engine:** MINT Intelligence & Guardrail Core (Real-Time Market Scanning, Strategy Generator, Hard-Cap Guardrail Engine, NLP Intent Parser)
- **Stellar Layer:** Integration & Execution (Horizon API / Soroban RPC, Key Management System, Path Payment Engine, SDEX Orderbook Execution)
- **Network Core:** Stellar Protocols (SDEX Orderbook & Liquidity Pools, Stellar Anchors via SEP-12/SEP-24 Fiat On/Off Ramps)

4. Key Functional Features

4.1. AI Market Intelligence & Autonomous Trading

- **Real-time Technical Analysis:** Analyzes price trends, DEX orderbook depth, liquidity pools, and volume changes on Stellar.
- **Strategy Preparation:** Generates trade proposals (e.g., limit orders, arbitrage, automated rebalancing) complete with rationale, risk rating, expected return, and maximum slippage.
- **Execution Modes:**

    - **Copilot Mode:** MINT suggests trades; the user confirms with one click.
    - **Autonomous Mode:** MINT executes trades automatically within pre-configured rules and hard caps.

4.2. Risk Mitigation & Hard Caps (Guardrails)

- **Hard Capital Cap:** Absolute dollar/asset limit MINT is authorized to manage (e.g., maximum $500 allocated to autonomous trading).
- **Daily Loss Limit:** Maximum allowed drawdown per 24 hours (e.g., 5% max loss). If reached, trading is automatically halted, orders are canceled, and a user alert is issued.
- **Max Per-Trade Size:** Limits the percentage of allocated funds used in a single order (e.g., no single trade exceeds 10% of the active trading pool).
- **Non-Custodial Multi-Sig Controls:** High-value trades or cap adjustments require secondary authorization or hardware wallet confirmation.

4.3. Stellar DEX Integration

- **Direct Orderbook Trading:** Submits passive and active limit orders directly to the SDEX.
- **Path Payments:** Uses Stellar's native path-finding algorithms to execute token swaps through multiple orderbooks or liquidity pools to get the best execution rate.
- **Asset Support:** Native XLM, major stablecoins (USDC, EURC), and trusted Stellar Anchor-issued tokens.

4.4. Payments & Money Transfers (Add-on Modules)

- **Direct P2P Transfer:** Send money using human-readable Stellar handles, federation addresses, or QR codes.
- **Cross-Currency Payments:** Send Asset A (e.g., XLM) and deliver Asset B (e.g., USDC) seamlessly via path payments.
- **Fiat On/Off Ramps:** Seamless integration with Stellar Anchors for local bank deposits and withdrawals.
- **Scheduled & Recurring Payments:** Set up automated payouts (e.g., monthly subscriptions, allowance) driven by smart triggers or simple schedules.

5. User Workflows

5.1. AI-Driven Trade Execution

1. **Parameters Configuration:** User sets risk parameters in MINT (e.g., Active Capital: $1,000 | Daily Max Loss: 3% | Max Trade Size: $100).
2. **Opportunity Detection:** MINT detects a price mismatch or arbitrage path on SDEX (e.g., XLM/USDC vs. XLM/yXLM/USDC).
3. **Proposal Generation:** MINT generates trade order (e.g., Buy 500 XLM at 0.12 USDC).

4. **Guardrail Verification:** MINT execution engine verifies trade against Hard Cap rules:
    - **Pass:** Order signed and broadcasted via Stellar RPC.
    - **Fail:** Order rejected; user alerted of threshold breach.

5.2. Natural Language Payment Transfer

1. **Intent Prompt:** User prompts: *"Send $25 USDC to @alex from my balance."*
2. **Parsing & Staging:** MINT parses intent, checks account balance, resolves recipient address, and generates transaction preview.
3. **Authentication:** User verifies with biometrics (Passkey/FaceID).
4. **Settlement:** Transaction settles on the Stellar network within 3–5 seconds.

6. Non-Functional Requirements

- **Performance:** Transaction finality within 3 to 5 seconds (Stellar network standard). MINT AI analysis latency < 2 seconds.
- **Security:** Non-custodial key storage using secure enclaves, Passkeys (WebAuthn), or Soroban smart contracts.
- **Scalability:** API throughput capable of processing 1,000+ concurrent market parsing requests per second.
- **Compliance:** KYC/AML integration for fiat on/off-ramp pathways via Stellar Anchor standards (SEP-12, SEP-24).

7. Implementation Roadmap & Milestones

- **Phase 1: Foundation (Q1)** - Non-custodial wallet auth, Basic P2P payments, SEP Anchor Fiat Ramps
- **Phase 2: Trading Engine (Q2)** - SDEX Orderbook integration, Path Payment routing engine, Automated trade execution
- **Phase 3: AI & Guardrails (Q3)** - Risk guardrail system, Autonomous MINT AI agent integration, Strategy testing simulator